SEQUENTIAL REALIZABILITY OF A FLAT VERTEX: A RIGID LAYERED MODEL AND ONE SIX-CREASE CONFIGURATION

ABSTRACT. We fix a combinatorial model of folding a single flat vertex by a sequence of steps, each of which folds along one line of the current folded image and may create and erase creases. We add a layered refinement in which each step is a rigid half-turn of a block of layers. For the vertex $A = \{0, 1, 3, 4, 5, 7\} \cdot \pi/4$ we show that a natural history with cancellation, namely *fold both diagonals, fold the full cross, unfold the line* $\{\pi/2, 3\pi/2\}$, is valid in the base model and passes the collision test (C), but admits no rigid realization. We introduce the *surviving partition* of a history, which groups the final creases by the step that last created them and forgets the order of the steps. We conjecture that no rigidly realizable history of A has the surviving partition $\{\{0, 4\}, \{1, 5\}, \{3, 7\}\}$ (three straight lines), nor indeed any surviving partition into straight lines through the vertex.

1. BASE MODEL

Conventions. $\mathbb{T} := \mathbb{R}/2\pi\mathbb{Z}$ (directions), ordered via representatives in $[0, 2\pi)$; $\mathbb{P} := \mathbb{R}/\pi\mathbb{Z}$ (lines) with projection $q\colon \mathbb{T} \to \mathbb{P}$. (a, b) denotes the open counterclockwise arc from a to b, and cl its closure. For $c \in \mathbb{T}$ let $\sigma_c(x) := 2c - x$. Since $2(c + \pi) = 2c$ in $\mathbb{T}$, σ_c depends only on $q(c)$, so σ_ℓ is defined for $\ell \in \mathbb{P}$. $\mathrm{Iso}(\mathbb{T}) := \{x \mapsto \varepsilon x + t : \varepsilon = \pm 1,\ t \in \mathbb{T}\}$.

Fact 1.1. For $g \in \mathrm{Iso}(\mathbb{T})$ and $x \in \mathbb{T}$: $g \circ \sigma_x \circ g^{-1} = \sigma_{g(x)}$.

Proof. For $g(y) = \varepsilon y + t$: $g\sigma_x g^{-1}(y) = \varepsilon\big(2x - \varepsilon(y - t)\big) + t = 2(\varepsilon x + t) - y$. □

Definition 1.2 (Faces, folding maps, flatness)**.** Let $S \subset \mathbb{T}$ be finite, $S = \{s_1 < \cdots < s_m\}$. If $m \geq 1$ the *faces* of S are $F_j := (s_j, s_{j+1})$ (indices mod m); if $S = \emptyset$ the only face is $\mathbb{T}$. The *folding maps* are $\varphi_1 := \mathrm{id}$ and $\varphi_{j+1} := \varphi_j \circ \sigma_{s_{j+1}}$ for $1 \leq j < m$. S is *flat* if $S = \emptyset$ or $\varphi_m \circ \sigma_{s_1} = \mathrm{id}$.

Fact 1.3. Let $S \neq \emptyset$ be flat. Then m is even and every face has length $\leq \pi$ (Kawasaki). The map $\Phi_S\colon \mathbb{T} \to \mathbb{T}$, $\Phi_S|_{\mathrm{cl}\, F_j} := \varphi_j$, is well defined and continuous, since $\varphi_{j+1}(s_{j+1}) = \varphi_j\sigma_{s_{j+1}}(s_{j+1}) = \varphi_j(s_{j+1})$ and, at s_1, $\varphi_m(s_1) = \varphi_m\sigma_{s_1}(s_1) = s_1 = \varphi_1(s_1)$. Set $\Phi_\emptyset := \mathrm{id}$.

Definition 1.4. For flat S and $\ell \in \mathbb{P}$ let $\Pi_S(\ell) := \{x \in \mathbb{T} : q(\Phi_S(x)) = \ell\}$. It is finite: Φ_S is an isometry on each closed face, which has length $\leq \pi$, so each closed face contains at most two points of $\Pi_S(\ell)$; and $\Pi_\emptyset(\ell) = q^{-1}(\ell)$.

Definition 1.5 (Step)**.** A *step* from a flat S is a nonempty $D \subset \mathbb{T}$ with $|D|$ even and $D \subseteq \Pi_S(\ell)$ for some $\ell \in \mathbb{P}$; this ℓ is unique and denoted $\ell(D)$. Put $N(D) := D \setminus S$ (created), $K(D) := D \cap S$ (erased). The result is $S \triangle D$. This is exactly the original definition: (P) is $|D|$ even, and (L) is $D \subseteq \Pi_S(\ell)$.

Definition 1.6 (Pieces and labelings)**.** The *pieces* of a step D from S are the faces of $S \cup D$. A *labeling* is a map $\mu\colon \{\text{pieces}\} \to \{0, 1\}$ such that two consecutive pieces receive different labels iff their common endpoint lies in D. As $|D|$ is even there are exactly two labelings, and they are complementary. Label 1 means *moving*. The parent face of a piece p, i.e. the face of S containing p, is denoted $\pi(p)$.

Lemma 1.7. *Let D be a step from flat S, $\ell = \ell(D)$, and μ a labeling. Define $\Psi_\mu := \Phi_S$ on closures of pieces with $\mu = 0$ and $\Psi_\mu := \sigma_\ell \circ \Phi_S$ on closures of pieces with $\mu = 1$. Then $S \triangle D$ is flat, and $\Phi_{S\triangle D} = g \circ \Psi_\mu$ for some $g \in \mathrm{Iso}(\mathbb{T})$.*

Proof. Ψ_μ is well defined and continuous because σ_ℓ fixes $\Phi_S(D) \subseteq q^{-1}(\ell)$ pointwise. Compare the restrictions of Ψ_μ to the two pieces meeting at a point $x \in S \cup D$. Below we treat the case where the piece before x has label 0; the other case follows by composing both restrictions on the left with σ_ℓ. If $S = \emptyset$, read $\varphi_j = \mathrm{id}$.

- If $x \in S \setminus D$, $x = s_{j+1}$, the labels agree and the restrictions are φ_j and $\varphi_{j+1} = \varphi_j \sigma_x$.
- If $x \in N(D)$ lies in a face F_j, then $q(\varphi_j(x)) = \ell$, so by Fact 1.1 $\varphi_j \sigma_x = \sigma_{\varphi_j(x)} \varphi_j = \sigma_\ell \varphi_j$. So the restrictions are φ_j and $\sigma_\ell \varphi_j = \varphi_j \sigma_x$, and x becomes a fold.
- If $x \in K(D)$, $x = s_{j+1}$, then the restrictions are φ_j and $\sigma_\ell \varphi_{j+1} = \sigma_\ell \varphi_j \sigma_x = \sigma_\ell \sigma_\ell \varphi_j = \varphi_j$, using $\varphi_j \sigma_x = \sigma_\ell \varphi_j$ as before. So they coincide and the fold at x disappears.

Thus Ψ_μ is continuous, piecewise in $\mathrm{Iso}(\mathbb{T})$, and changes by σ_x exactly at the points x of $S \triangle D$. Let $g \in \mathrm{Iso}(\mathbb{T})$ make $g \circ \Psi_\mu$ the identity on the first face of $S \triangle D$. Then the folding maps of $S \triangle D$ are the restrictions of $g \circ \Psi_\mu$, and the closing condition holds because $g \circ \Psi_\mu$ is continuous around the whole circle. (If $S \triangle D = \emptyset$, Ψ_μ is a single isometry and $g = \Psi_\mu^{-1}$.) □

Lemma 1.8 (Reversibility)**.** *If D is a step from S, then D is a step from $S \triangle D$.*

Proof. For $d \in D$, $\Psi_\mu(d) = \Phi_S(d)$ because σ_ℓ fixes $q^{-1}(\ell)$ pointwise. By Lemma 1.7, $\Phi_{S\triangle D}(d) = g(\Phi_S(d)) \in g\big(q^{-1}(\ell)\big)$, a single line, for all $d \in D$. □

Definition 1.9 (Histories)**.** A *history* is a tuple $H = (D_1, \dots, D_n)$ with $S_0 = \emptyset$, $S_i = S_{i-1} \triangle D_i$, and each D_i a step from S_{i-1}. It is a history of A if $S_n = A$. For $a \in A$ let $\tau_H(a) := \max\{i : a \in N(D_i)\}$; the set is nonempty since $a \notin S_0$. The *structure* $\mathrm{struct}(H)$ is the preorder $a \preceq b \iff \tau_H(a) \le \tau_H(b)$. The *cost* is $\mathrm{cost}(H) := \sum_i |K(D_i)|$. $R(A)$ is the set of structures of histories of A, and $R_0(A)$ the subset coming from histories of cost 0.

Definition 1.10 (Surviving partition, straight blocks)**.** The *surviving partition* of a history $H = (D_1, \dots, D_n)$ of A is

$$\mathcal{P}(H) := \big\{\tau_H^{-1}(i) : 1 \le i \le n\big\} \setminus \{\emptyset\},$$

a partition of A. Its blocks are called the *surviving blocks* of H: the block $\tau_H^{-1}(i)$ is what survives in A of the creations made by step i. A set $B \subseteq \mathbb{T}$ is *straight* if $B + \pi = B$, i.e. B is a union of lines through the vertex.

Remark 1.11. $\mathcal{P}(H)$ is the partition underlying $\mathrm{struct}(H)$, with the order of the classes forgotten. It ignores scaffold creases, i.e. creases created and later erased, and it ignores earlier creations of creases of A that were erased and created again: only the last creation counts.

Definition 1.12 (Collision condition (C))**.** A step D from S satisfies (C) if the following holds. Whenever F, F' are faces of S with $F \cap D = F' \cap D = \emptyset$, and $x \in F$, $x' \in F'$ satisfy $\Phi_S(x) = \Phi_S(x') \in q^{-1}(\ell(D))$, then $\mu(F) = \mu(F')$. Such faces are pieces, and the condition does not depend on which labeling is used.

2. Layered model

Definition 2.1 (Layered state)**.** Let S be flat. For a face F put $W_F := \Phi_S(F)$, an open arc (or $\mathbb{T}$ if $S = \emptyset$). Let $E_S := \Phi_S(S)$, a finite set, and for $u \notin E_S$ let $L(u) := \{F : u \in W_F\}$. A *layer order* on S is a family $(<_u)_{u \in \mathbb{T} \setminus E_S}$ of strict total orders on $L(u)$ that is locally constant in u.

For $v \in E_S$, write $<_{v^+}$ and $<_{v^-}$ for the order at points of $\mathbb{T} \setminus E_S$ immediately counterclockwise, resp. clockwise, of v; likewise $L(v^\pm)$. Put $\Theta(v) := L(v^-) \cap L(v^+)$, the faces *passing through*

v. The *taco* of a ray $s = s_{j+1} \in S$ is $\{F_j, F_{j+1}\}$. Since $\varphi_{j+1}(s+t) = \varphi_j(s-t)$, both faces of the taco of s lie in $L(v^\epsilon)$ for exactly one sign $\epsilon = \epsilon(s)$, where $v = \Phi_S(s)$. The pair $(S, <)$ is a *layered state* if the following three conditions hold.

(O1) For every $v \in E_S$, the orders $<_{v^-}$ and $<_{v^+}$ agree on $\Theta(v)$.
(O2) If $s \neq s'$, $\Phi_S(s) = \Phi_S(s') = v$ and $\epsilon(s) = \epsilon(s') = \epsilon$, then the two tacos, which are disjoint, do not interleave in $<_{v^\epsilon}$: it is not the case that exactly one face of one taco lies strictly between the two faces of the other.
(O3) If $v = \Phi_S(s)$ and $c \in \Theta(v)$, then c does not lie strictly between the two faces of the taco of s in $<_{v^{\epsilon(s)}}$.

The *initial layered state* is $\emptyset$ with its unique layer order.

Definition 2.2 (Rigid step)**.** The data are a layered state $(S, <)$, a step D with $\ell = \ell(D)$, a labeling μ, and a choice of one component $H^\uparrow$ of $\mathbb{T} \setminus q^{-1}(\ell)$; the other component is $H^\downarrow$. Physically: a half-turn about ℓ that lifts the layers over $H^\uparrow$ and lowers those over $H^\downarrow$.

(F) Let $u \notin E_S \cup q^{-1}(\ell)$, and let p, p' be pieces with $\mu(p) = 1$, $\mu(p') = 0$ and $u \in \Phi_S(p) \cap \Phi_S(p')$. Then $\pi(p') <_u \pi(p)$ if $u \in H^\uparrow$, and $\pi(p) <_u \pi(p')$ if $u \in H^\downarrow$.

New order. Let $S' = S \triangle D$ and let $\Psi = \Psi_\mu$ and g be as in Lemma 1.7. Each face G of S' is a union of pieces and points of $K(D)$, and Ψ is injective on G. Let u avoid the finite set $E_S \cup \sigma_\ell(E_S) \cup q^{-1}(\ell)$. Define $\prec_u$ on the pieces p with $u \in \Psi(p)$ as follows:

(i) two staying pieces: $p \prec_u p' \iff \pi(p) <_u \pi(p')$;
(ii) two moving pieces: $p \prec_u p' \iff \pi(p') <_{\sigma_\ell(u)} \pi(p)$, i.e. the order is reversed;
(iii) p moving and p' staying: $p' \prec_u p$ if $u \in H^\downarrow$, and $p \prec_u p'$ if $u \in H^\uparrow$.

Set $G <'_{g(u)} G'$ iff $p_G \prec_u p_{G'}$, where p_G is the unique piece of G with $u \in \Psi(p_G)$. The step is *rigid* if (F) holds and $<'$ extends to a layer order making $(S', <')$ a layered state.

A history is *rigidly realizable* if there are layered states $(S_i, <^i)$, starting from the initial one, such that each transition $(S_{i-1}, <^{i-1}) \to (S_i, <^i)$ is a rigid step for D_i, for some labeling and some choice of $H^\uparrow$. Let

$$\mathcal{P}_{\mathrm{rig}}(A) := \{\mathcal{P}(H) : H \text{ a rigidly realizable history of } A\}$$

be the set of *rigidly realizable surviving partitions* of A.

Lemma 2.3. *(F) implies (C). Consequently every step of a rigidly realizable history satisfies (C).*

Proof. Let F, F', x, x' be as in Definition 1.12 with $w := \Phi_S(x) = \Phi_S(x')$, and suppose $\mu(F) = 1 \neq 0 = \mu(F')$. Since x, x' are interior points of faces, $F, F' \in \Theta(w)$ if $w \in E_S$, and otherwise $F, F' \in L(u)$ for all u near w on both sides; in both cases their relative order is the same on both sides of w (by (O1), resp. local constancy). But (F) requires $\pi(F') <_u \pi(F)$ for $u \in H^\uparrow$ near w and $\pi(F) <_u \pi(F')$ for $u \in H^\downarrow$ near w. Contradiction. □

Lemma 2.4 (Simple folds)**.** *Let $D = \Pi_S(\ell)$ with $D \cap S = \emptyset$. Then the image of each piece lies in one component of $\mathbb{T} \setminus q^{-1}(\ell)$, and each labeling moves exactly the pieces lying over one component. Hence (F) holds vacuously, and for each of the four choices of labeling and $H^\uparrow$ the order $<'$ is determined by (i)–(iii).*

Proof. No ray maps into $q^{-1}(\ell)$, since $\Pi_S(\ell) \cap S = D \cap S = \emptyset$, and all interior points of faces mapping into $q^{-1}(\ell)$ lie in D. So no piece's image meets $q^{-1}(\ell)$ in its interior. Across a point of D (a transversal crossing) both the side and the label switch; across a ray of S (a fold away from ℓ) neither switches. □

3. The configuration $\{0,1,3,4,5,7\}\cdot\pi/4$

From now on the unit is $\pi/4$, so $\mathbb{T}=\mathbb{R}/8\mathbb{Z}$, and

$$A=\{0,1,3,4,5,7\}.$$

Let $f_k:=(k-1,k)$, $k=1,\dots,8$, be the faces of $\{0,\dots,7\}$. In the drawing convention where direction 0 points up, $\{0,4\}$ is the vertical and $\{2,6\}$ the horizontal line. The three lines of A are $\{0,4\}$, $\{1,5\}$ and $\{3,7\}$, and the straight subsets of A are exactly the unions of these. Consider

$$\begin{aligned}H_1&=\big(\{0,4\},\ \{1,7\},\ \{3,5\}\big),\\ H_2&=\big(\{1,5\},\ \{3,7\},\ \{0,2,4,6\},\ \{2,6\}\big),\\ H_3&=\big(\{1,5\},\ \{3,7\},\ \{0,4\}\big).\end{aligned}$$

Proposition 3.1.

(a) H_1,H_2,H_3 *are histories of* A.
(b) $\mathcal{P}(H_1)=\{\{0,4\},\{1,7\},\{3,5\}\}$*: one straight block and two non-straight blocks, which are mirror images of each other across the line* $\{0,4\}$ *(since* $7=-1$, $5=-3$*).* $\mathcal{P}(H_2)=\mathcal{P}(H_3)=\Lambda:=\{\{0,4\},\{1,5\},\{3,7\}\}$*: three straight blocks.*
(c) $\operatorname{cost}(H_1)=\operatorname{cost}(H_3)=0$ *and* $\operatorname{cost}(H_2)=2$*. In particular* Λ *is the surviving partition of a history of cost* 0*, namely* H_3.
(d) Every step of H_1 *and* H_2 *satisfies (C). Step* 3 *of* H_3 *violates (C).*

Proof. We record the normalised folding maps (Definition 1.2) face by face.

H_1. $\Pi_\emptyset(q(0))=\{0,4\}=D_1$. For $S_1=\{0,4\}$: $\Phi=x$ on $(0,4)$ and $-x$ on $(4,8)$, so $\Pi_{S_1}(q(1))=\{1,7\}=D_2$. For $S_2=\{0,1,4,7\}$, Φ is x, $2-x$, $x+2$, $-x$ on $(0,1),(1,4),(4,7),(7,8)$ respectively, so $\Phi(3)=\Phi(5)=7$ and $\Pi_{S_2}(q(7))=\{3,5\}=D_3$.

H_2,H_3. $\Pi_\emptyset(q(1))=\{1,5\}=D_1$. For $S_1=\{1,5\}$: $\Phi=x$ on $(1,5)$ and $2-x$ on $(5,9)$, so $\Pi_{S_1}(q(3))=\{3,7\}=D_2$. For $S_2=\{1,3,5,7\}$: $\Phi=x,\ 6-x,\ x-4,\ 2-x$ on $(1,3),(3,5),(5,7),(7,9)$; rays map to $1,3,1,3$, and $\Phi(0)=\Phi(2)=\Phi(4)=\Phi(6)=2$, so $\Pi_{S_2}(q(2))=\{0,2,4,6\}$. Hence D_3 is a step for H_2, and so is $\{0,4\}$ for H_3. For $S_3=\{0,\dots,7\}$ in H_2: $\Phi(k)=0$ for even k and $\Phi(k)=1$ for odd k, with all faces mapping onto $(0,1)$, so $\Pi_{S_3}(q(0))=\{0,2,4,6\}\supseteq D_4$.

(b) and (c) are read off directly. For (d): in H_1 and in steps 1–3 of H_2 we have $D=\Pi_S(\ell)$, so no face avoiding D contains a preimage of ℓ. In step 4 of H_2, $\Pi_{S_3}(q(0))$ consists of rays, so no face contains one in its interior. In step 3 of H_3, the faces $(1,3)$ and $(5,7)$ of S_2 avoid $\{0,4\}$ and contain 2 and 6, with $\Phi(2)=\Phi(6)=2\in q^{-1}(q(2))$; the labeling switches at 0 and 4, so these faces receive opposite labels. □

Theorem 3.2. H_2 *is not rigidly realizable.*

Proof. Steps 1–3 of H_2 satisfy the hypothesis of Lemma 2.4, so each admits exactly four candidate realizations (two labelings times two choices of $H^{\uparrow}$), each determined by (i)–(iii). Running all $4^3=64$ combinations from the initial layered state (Appendix A) yields, after step 3, eight faces $f_1,\dots,f_8$ that all have the same image arc W (up to the global isometry g), of length 1 with an endpoint in $q^{-1}(\ell_4)$. Their order $<_W$, bottom to top, is one of the following eight; the second column is the reversal of the first.

$f_2f_1f_6f_5\,f_4f_7f_8f_3$	$f_3f_8f_7f_4\,f_5f_6f_1f_2$
$f_4f_7f_8f_3\,f_2f_1f_6f_5$	$f_5f_6f_1f_2\,f_3f_8f_7f_4$
$f_6f_5f_2f_1\,f_8f_3f_4f_7$	$f_7f_4f_3f_8\,f_1f_2f_5f_6$
$f_8f_3f_4f_7\,f_6f_5f_2f_1$	$f_1f_2f_5f_6\,f_7f_4f_3f_8$

In step 4 ($D_4 = \{2, 6\}$) the two labelings move $M = \{f_3, f_4, f_5, f_6\}$ or its complement. Since W lies in a single component of $\mathbb{T} \setminus q^{-1}(\ell_4)$, condition (F) at any $u \in W$ says that the moving set is an up-set of $<_W$ (if $W \subseteq H^{\uparrow}$) or a down-set (if $W \subseteq H^{\downarrow}$). In each of the eight orders, neither M nor its complement is the set of the four lowest faces. Hence (F) fails for every choice. □

Remark 3.3. Theorem 3.2 does not depend on whether simple folds preserve (O1)–(O3): if they do not, H_2 has no realization at all. It shows that layer data are needed, since H_2 passes (C) by Proposition 3.1(d). We expect H_1, a sequence of three simple folds, to be rigidly realizable, but have not written out the (O1)–(O3) check.

Conjecture 3.4 (Three lines)**.** $\Lambda \notin \mathcal{P}_{\mathrm{rig}}(A)$. *That is, no rigidly realizable history of A, with any number of scaffold creases and in any order of steps, has the surviving blocks* $\{0, 4\}$, $\{1, 5\}$, $\{3, 7\}$.

Conjecture 3.5 (No straight surviving partition)**.** *No partition in $\mathcal{P}_{\mathrm{rig}}(A)$ has all its blocks straight.*

Remark 3.6.

(1) *Relation.* Conjecture 3.5 implies Conjecture 3.4. The partitions of A into straight blocks are the five partitions of the set of its three lines: Λ, the three partitions merging two lines, and $\{A\}$.
(2) *Order is free.* Both conjectures constrain only $\mathcal{P}(H)$, so they cover every order in which the blocks may be finally created, e.g. $\{0, 4\}$ first, in the middle, or last.
(3) *Symmetry.* The isometries $x \mapsto -x$, $x \mapsto 4 - x$ and $x \mapsto x + 4$ preserve A and permute its three lines. The model is equivariant under $\mathrm{Iso}(\mathbb{T})$ (the normalisation in Definition 1.2 only changes images by a global isometry), so both conjectures are invariant under these symmetries.
(4) *Evidence.* H_2 and H_3, the two histories with $\mathcal{P} = \Lambda$ exhibited here, are not rigidly realizable: H_2 by Theorem 3.2, and H_3 because its third step violates (C), hence (F) by Lemma 2.3. By contrast H_1, which we expect to be rigidly realizable, has only one straight block.
(5) *Status.* Histories with a given surviving partition are unbounded in length, so neither conjecture is a finite check. A counterexample, if one exists, can be found by enumerating combinatorial history types.
(6) *Rigidity is essential.* Suppose instead that creases persist once made and fold angles vary continuously, so that unfolding a crease to angle π does not erase it. Then Λ is realized trivially: fold $\{1, 5\}$, fold $\{3, 7\}$, unfold both to the flat sheet, fold $\{0, 4\}$, and refold the existing creases as $\{1, 7\}$ and $\{3, 5\}$, as in H_1. Each crease of A is created exactly once, in the blocks $\{1, 5\}$, $\{3, 7\}$, $\{0, 4\}$. More generally, connectedness results for single-vertex configuration spaces (Streinu–Whiteley, *Single-vertex origami and spherical expansive motions*, JCDCG 2004, LNCS 3742; exact hypotheses not re-checked) suggest that any folded state is then reachable.

Appendix A. Verification script for Theorem 3.2

Unit arcs $(a, a + 1)$ of the image carry stacks of faces (bottom to top). A fold through the image point c moves the arcs on one side, reflects them via $a \mapsto 2c - a - 1$, reverses each moved stack, and places it on top (over) or at the bottom (under) of the stack at the landing arc. The fold point of each step is the current image of a crease point of that step. In `choices`, `side` selects the labeling (which side moves) and `over` selects $H^{\uparrow}$: the moving side is $H^{\uparrow}$ iff `over`, matching rule (iii).

```
import itertools
def run(choices):
```

```
    g = {k: (1, 0) for k in range(1, 9)}        # image map of f_k: x -> e*x+t (mod 8)
    stack = {a: [a+1] for a in range(8)}        # unit arc (a,a+1) -> faces, bottom->top
    img = lambda k, x: (g[k][0]*x + g[k][1]) % 8
    for (pt, face), (side, over) in zip([(1,1), (3,3), (0,1)], choices):
        c = img(face, pt)                       # fold line through image of a crease point
        plus = lambda a: ((a - c) % 8) < 4
        mov = {a for a in range(8) if plus(a) == (side == 0)}
        new = {a: list(stack[a]) for a in range(8) if a not in mov}
        for a in mov:
            b = (2*c - a - 1) % 8
            r = list(reversed(stack[a])); new.setdefault(b, [])
            new[b] = new[b] + r if over else r + new[b]
        for a in mov:
            for k in stack[a]:
                e, t = g[k]; g[k] = (-e, (2*c - t) % 8)
        stack = {a: new.get(a, []) for a in range(8)}
    return stack
M = {3, 4, 5, 6}
res = set()
for ch in itertools.product(itertools.product((0,1),(0,1)), repeat=3):
    st = run(ch); arcs = [a for a in st if st[a]]
    assert len(arcs) == 1 and len(st[arcs[0]]) == 8   # all faces share one image arc
    res.add(tuple(st[arcs[0]]))
for s in sorted(res):                                 # 8 orders, none admits a free block
    print(s, set(s[:4]) in (M, set(range(1, 9)) - M))
```

The output consists of the eight orders listed in the proof of Theorem 3.2, each with the value `False`.